Niels Christoffersen

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Education

Reed College **Aug. 2020 – May 2024**
Bachelor of Arts in Computer Science *Portland, Oregon*

Technical Skills

Machine Learning: TensorFlow, PyTorch, Keras, Scikit-learn, Deep Learning (CNNs, RNNs, Transformers), Model Deployment
Languages: Python, C++, C#, SQL, Bash, R, JavaScript/Node.js, Scala
ML Operations: Docker, Kubernetes, MLflow, Model Optimization, A/B Testing, Production Pipelines
Data Engineering: Apache Spark, Cassandra, PostgreSQL, ETL Pipelines, Large-Scale Data Processing
Cloud & Tools: AWS, Git, Linux, REST APIs, CI/CD, Distributed Systems

Relevant Coursework

- Deep Learning
- Linear Algebra
- Algorithms & Data Structures
- Database Systems
- Computer Systems
- Private and Fair Data Analysis
- Computability & Complexity
- Philosophy of AI
- Cryptography

Experience

Freelance Software Developer **2024 – Present**
Self-Employed *Remote*

- Develop Python automation scripts and data processing tools for accounting clients, building custom solutions for financial data analysis and workflow automation
- Design and implement production scripts handling large datasets with automated reporting, data validation, and error handling for reliable daily operations

Los Alamos National Laboratory **May 2021 – July 2021**
Machine Learning & Software Engineering Intern *Los Alamos, New Mexico*

- Developed and optimized Python-based data processing pipelines handling terabyte-scale datasets for satellite constellation modeling and nuclear event detection analytics, processing millions of data points with high throughput and reliability
- Built production-grade infrastructure for real-time space vehicle data analysis with strict uptime requirements, implementing automated testing, data validation, and continuous integration workflows using bash scripting and Linux command-line tools
- Engineered scalable build system migration for the DIORAMA Framework, achieving 50% compilation time reduction and improved deployment efficiency for mission-critical simulation systems
- Collaborated cross-functionally with data science and DevOps teams to maintain data quality standards and ensure system reliability for U.S. Space Nuclear Detonation Detection program

Projects

Deep Learning Sports Analytics Platform | *TensorFlow, PyTorch, Docker* **Nov 2024 – Present**

- Trained and deployed LSTM deep learning model using Pytorch into production with measurably improved accuracy over baseline, processing 10+ years of sports data with millions of records
- Built production REST API with Docker containerization serving real-time model predictions with high throughput and 99.9% uptime
- Engineered scalable ML pipeline with Python, SQL, and bash for automated data collection, feature engineering, training, and deployment with hyperparameter tuning and continuous improvement
- Open-sourced at github.com/nielsjsc/LSTMLB — Live demo at longballhq.xyz

MusicBeeWrapped Plugin | *C#, .NET, Data Analytics* **2024**

- Built Spotify Wrapped alternative as a plugin for MusicBee music player which tracks listening behavior, analyzes patterns, and generates yearly reviews while keeping all data on-device for complete privacy
- Designed data pipeline that continuously logs play events with smart filtering (5-second minimum to ignore skips), handles multi-year storage with automatic schema migrations, and processes millions of listening sessions
- Created automated report generator that transforms raw listening data into interactive web-based insights, all compiled and served locally without external dependencies

Additional Information

Location: Bellevue, WA (Greater Seattle Area)
Interests: Staying current with latest deep learning research, contributing to open-source ML projects, scalable systems